Customer Churn Analysis Project Report

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This report presents a business-friendly overview of the Customer Churn Analysis project. The objective was to identify factors influencing customer attrition and provide actionable recommendations to reduce churn and protect revenue. The analysis was performed using Python, SQL, and Power BI.

Executive Summary

Dataset size: 7,043 customer records.
Observed churn rate: 26.54% (based on the source dataset).
Key business finding: Customers on month-to-month contracts, customers with certain payment methods, and customers with fewer subscribed services are more likely to churn.

Project Objectives

Measure customer churn and its business impact. Identify customer segments with the highest churn risk. Analyze contract type, tenure, payment method, internet service, and demographics. Provide actionable recommendations for retention. Create an interactive Power BI dashboard for decision-makers.

Technologies Used

Python: Data cleaning, transformation, and exploratory analysis.
Pandas: Data manipulation and aggregation.
NumPy: Numerical operations and feature preparation.
Matplotlib & Seaborn: Visualizations and trend analysis.
SQL: Data extraction, filtering, aggregation, and business queries.
Power BI: Interactive dashboard development and KPI reporting.

Methodology

1. Data Collection – Imported the Telco Customer Churn dataset.
2. Data Cleaning – Handled missing values, corrected data types, and standardized fields.
3. Exploratory Data Analysis – Investigated churn behavior across customer segments.
4. SQL Analysis – Executed business queries to identify patterns and KPIs.
5. Dashboard Development – Built interactive Power BI visualizations and filters.
6. Recommendation Framework – Converted insights into business actions.

SQL Usage

SQL scripts were used to perform customer segmentation, aggregation, KPI calculations, and churn-related business analysis. The uploaded SQL file contains approximately 137 lines of SQL logic.

Power BI Dashboard

Figure 1: Main Customer Churn Analysis Dashboard

Figure 2: Deep Dive Analysis and Recommendations Dashboard

Key Findings

Month-to-month contracts show the highest churn risk. Longer-tenure customers are significantly more likely to stay. Certain payment methods are associated with higher churn. Customers with more subscribed services tend to be more loyal. Senior citizens represent a notable at-risk segment. Internet service type influences churn behavior.

Business Recommendations

Encourage migration from month-to-month to annual contracts through loyalty incentives. Implement a structured onboarding program for new customers. Promote auto-pay enrollment and frictionless billing. Investigate service-quality concerns in high-churn customer groups. Develop targeted retention campaigns for high-risk segments. Track churn KPIs continuously through the Power BI dashboard.

Conclusion

This project combines Python analytics, SQL-based business querying, and Power BI visualization to deliver actionable customer-retention insights. The dashboard enables stakeholders with non-technical backgrounds to understand churn drivers, monitor KPIs, and make informed decisions to reduce revenue loss and improve customer retention.